

```

Last login: Sun Jun 28 07:02:06 on ttys000
stephenrajva@Stephens-MacBook-Air-4 ~ % find "/Users/stephenrajva/Downloads/HPC-
Upload 2" -name "minimd_mesh3d_ppn1.py" 2>/dev/null
find "/Users/stephenrajva/Downloads/HPC-Upload 2" -name "minimd_mesh*ppn1*.py" 2
>/dev/null
ls "/Users/stephenrajva/Downloads/HPC-Upload 2/miniapps/MiniMD_simulation/mesh3D
/" 2>/dev/null | grep "\.py$"
/Users/stephenrajva/Downloads/HPC-Upload 2/miniapps/MiniMD_simulation/mesh3D/min
imd_mesh3d_ppn1.py
/Users/stephenrajva/Downloads/HPC-Upload 2/miniapps/MiniMD_simulation/mesh3D/min
imd_mesh3d_ppn1.py
analyze_minimd_mesh3d.py
hopcount.py
mesh3D.py
minimd_mesh3d_ppn1.py
stephenrajva@Stephens-MacBook-Air-4 ~ % cat "/Users/stephenrajva/Downloads/HPC-U
pload 2/miniapps/MiniMD_simulation/mesh3D/minimd_mesh3d_ppn1.py"
#!/usr/bin/env python
# =====
# minimd_mesh3d_ppn1.py - MiniMD on a 3D mesh, PPN (ranks per node) = 1
# =====
# Same idea as the MiniFE study, but:
#   * the topology is 3D (shape like "4x4x4")
#   * the workload is the MiniMD pattern (neighbor halo exchange + collectives)
#
# PPN = ranks / nodes = job_size / (routers * local_ports).
# We set job_size = nodes, so PPN is ALWAYS 1.
#
# USAGE (one shape per run, chosen with an environment variable):
#     export MESH_SHAPE="4x4x4"
#     sst minimd_mesh3d_ppn1.py
# The runner script loops over all the shapes for you.
# =====

import sst
import os
from sst.merlin.base import *
from sst.merlin.endpoint import *
from sst.merlin.interface import *
from sst.merlin.topology import *
from sst.ember import *

# ---- parameters -----
PPN = 1
shape = os.environ.get("MESH_SHAPE", "2x2x4") # 3 numbers: XxYxZ
nodes_per_router = 1
out_dir = os.environ.get("STATS_DIR", ".")

dims = [int(x) for x in shape.lower().split("x")]
num_routers = 1
for d in dims:
    num_routers *= d
num_nodes = num_routers * nodes_per_router
num_ranks = num_nodes * PPN # PPN=1 -> ranks == nodes

PlatformDefinition.setCurrentPlatform("firefly-defaults")

# ---- topology (3D mesh) -----

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topo                = topoMesh()
topo.shape          = shape
topo.local_ports    = nodes_per_router
topo.width          = "x".join(["1"] * len(dims)) # "1x1x1" for 3D
topo.link_latency   = "20ns"

# ---- router -----
router              = hr_router()
router.link_bw      = "12GB/s"
router.flit_size    = "16B"
router.xbar_bw      = "24GB/s"
router.input_latency = "20ns"
router.output_latency = "20ns"
router.input_buf_size = "16kB"
router.output_buf_size = "16kB"
router.num_vns      = 2
router.xbar_arb      = "merlin.xbar_arb_lru"
topo.router = router

# ---- NIC -----
nic                = ReorderLinkControl()
nic.link_bw        = "12GB/s"
nic.input_buf_size = "16kB"
nic.output_buf_size = "16kB"

system = System()
system.setTopology(topo)

# ---- MiniMD workload (same motifs as your TAU-profiled run) -----
motifs = [
    {"pattern": "AllPingPong", "params": "iterations=325 messageSize=20510 computeTime=476000"},
    {"pattern": "AllPingPong", "params": "iterations=325 messageSize=9000 computeTime=476000"},
    {"pattern": "AllPingPong", "params": "iterations=325 messageSize=20510 computeTime=476000"},
    {"pattern": "AllPingPong", "params": "iterations=325 messageSize=9000 computeTime=476000"},
    {"pattern": "Allreduce", "params": "count=1 iterations=40 compute=2451000"},
    {"pattern": "Barrier", "params": "iterations=15 compute=1392000"},
]

ep = EmberMPIJob(0, num_ranks)
ep.network_interface = nic
ep.addMotif("Init")
for m in motifs:
    ep.addMotif(f"{m['pattern']} {m['params']}")
ep.addMotif("Fini")
ep.nic.nic2host_lat = "100ns"

system.allocateNodes(ep, "linear")
system.build()

# ---- statistics -> one CSV per shape -----
csv_path = os.path.join(out_dir, f"minimd_mesh_{shape}_ppn{PPN}_nodes{num_nodes}_stats.csv")

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sst.setStatisticLoadLevel(10)
sst.setStatisticOutput("sst.statOutputCSV", {"filepath": csv_path, "separator":
", "})

sst.enableAllStatisticsForComponentType("merlin.hr_router")
sst.enableAllStatisticsForComponentType("merlin.linkcontrol")
sst.enableAllStatisticsForComponentType("ember.EmberEngine", {"MotifLog": "1"})
sst.enableAllStatisticsForComponentType("ember.nic")

# ---- summary print (verify PPN at a glance) -----
print("=====")
print(f" Topology          : 3D Mesh {shape}")
print(f" Routers            : {num_routers}")
print(f" Total nodes         : {num_nodes}")
print(f" Total MPI ranks     : {num_ranks}")
print(f" PPN (ranks/node)    : {num_ranks // num_nodes}    <-- should be 1")
print(f" Stats CSV           : {csv_path}")
print("=====")
stephenrajva@Stephens-MacBook-Air-4 ~ % echo "==== PPN=1 mesh config ====="
cat "/Users/stephenrajva/Downloads/HPC-Upload 2/miniapps/MiniMD_simulation/mesh3
D/minimd_mesh3d_ppn1.py"
echo ""
echo "==== source folder (miniapps mesh3D) ====="
ls -la "/Users/stephenrajva/Downloads/HPC-Upload 2/miniapps/MiniMD_simulation/me
sh3D/"
echo ""
echo "==== destination folder (SST_Stimulator MiniMD/Mesh) ====="
ls -la "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniMD/Mesh/"
==== PPN=1 mesh config =====
#!/usr/bin/env python
# =====
# minimd_mesh3d_ppn1.py - MiniMD on a 3D mesh, PPN (ranks per node) = 1
# =====
# Same idea as the MiniFE study, but:
# * the topology is 3D (shape like "4x4x4")
# * the workload is the MiniMD pattern (neighbor halo exchange + collectives)
#
# PPN = ranks / nodes = job_size / (routers * local_ports).
# We set job_size = nodes, so PPN is ALWAYS 1.
#
# USAGE (one shape per run, chosen with an environment variable):
#     export MESH_SHAPE="4x4x4"
#     sst minimd_mesh3d_ppn1.py
# The runner script loops over all the shapes for you.
# =====

import sst
import os
from sst.merlin.base import *
from sst.merlin.endpoint import *
from sst.merlin.interface import *
from sst.merlin.topology import *
from sst.ember import *

# ---- parameters -----
PPN          = 1
shape        = os.environ.get("MESH_SHAPE", "2x2x4")    # 3 numbers: XxYxZ
nodes_per_router = 1

```

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out_dir          = os.environ.get("STATS_DIR", ".")

dims             = [int(x) for x in shape.lower().split("x")]
num_routers      = 1
for d in dims:
    num_routers *= d
num_nodes        = num_routers * nodes_per_router
num_ranks        = num_nodes * PPN                      # PPN=1 -> ranks == nodes

PlatformDefinition.setCurrentPlatform("firefly-defaults")

# ---- topology (3D mesh) -----
topo             = topoMesh()
topo.shape       = shape
topo.local_ports = nodes_per_router
topo.width       = "x".join(["1"] * len(dims)) # "1x1x1" for 3D
topo.link_latency = "20ns"

# ---- router -----
router           = hr_router()
router.link_bw   = "12GB/s"
router.flit_size = "16B"
router.xbar_bw   = "24GB/s"
router.input_latency = "20ns"
router.output_latency = "20ns"
router.input_buf_size = "16kB"
router.output_buf_size = "16kB"
router.num_vns    = 2
router.xbar_arb   = "merlin.xbar_arb_lru"
topo.router = router

# ---- NIC -----
nic             = ReorderLinkControl()
nic.link_bw     = "12GB/s"
nic.input_buf_size = "16kB"
nic.output_buf_size = "16kB"

system = System()
system.setTopology(topo)

# ---- MiniMD workload (same motifs as your TAU-profiled run) -----
motifs = [
    {"pattern": "AllPingPong", "params": "iterations=325 messageSize=20510 compute=476000"},
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    {"pattern": "AllPingPong", "params": "iterations=325 messageSize=20510 compute=476000"},
    {"pattern": "AllPingPong", "params": "iterations=325 messageSize=9000 compute=476000"},
    {"pattern": "Allreduce", "params": "count=1 iterations=40 compute=2451000"},
    {"pattern": "Barrier", "params": "iterations=15 compute=1392000"},
]

ep = EmberMPIJob(0, num_ranks)
ep.network_interface = nic
ep.addMotif("Init")

```

```

for m in motifs:
    ep.addMotif(f"{m['pattern']} {m['params']}")
ep.addMotif("Fini")
ep.nic.nic2host_lat = "100ns"

system.allocateNodes(ep, "linear")
system.build()

# ---- statistics -> one CSV per shape -----
csv_path = os.path.join(out_dir, f"minimd_mesh_{shape}_ppn{PPN}_nodes{num_nodes}_stats.csv")

sst.setStatisticLoadLevel(10)
sst.setStatisticOutput("sst.statOutputCSV", {"filepath": csv_path, "separator":
", "})

sst.enableAllStatisticsForComponentType("merlin.hr_router")
sst.enableAllStatisticsForComponentType("merlin.linkcontrol")
sst.enableAllStatisticsForComponentType("ember.EmberEngine", {"MotifLog": "1"})
sst.enableAllStatisticsForComponentType("ember.nic")

# ---- summary print (verify PPN at a glance) -----
print("=====")
print(f" Topology          : 3D Mesh {shape}")
print(f" Routers            : {num_routers}")
print(f" Total nodes         : {num_nodes}")
print(f" Total MPI ranks     : {num_ranks}")
print(f" PPN (ranks/node)    : {num_ranks // num_nodes}    <-- should be 1")
print(f" Stats CSV           : {csv_path}")
print("=====")

===== source folder (miniapps mesh3D) =====
total 1680
drwxr-xr-x 16 stephenrajva staff 512 24 Jun 15:41 .
drwxr-xr-x 6 stephenrajva staff 192 13 Jun 17:09 ..
-rw-r--r--@ 1 stephenrajva staff 8196 25 Jun 00:49 .DS_Store
-rw-r--r--@ 1 stephenrajva staff 4131 24 Jun 15:33 analyze_minimd_mesh3d.py
-rw-r--r-- 1 stephenrajva staff 4828 13 Jun 17:09 hopcount.py
-rw-r--r--@ 1 stephenrajva staff 2872 13 Jun 17:09 mesh3D.py
-rw-r--r-- 1 stephenrajva staff 2091 13 Jun 17:09 minimd_3dmesh_hopcount_s
ummary.csv
-rw-r--r-- 1 stephenrajva staff 26072 13 Jun 17:09 minimd_3dmesh_hopcount_s
ummary.csv.xlsx
-rw-r--r-- 1 stephenrajva staff 1315 13 Jun 17:09 minimd_3dmesh_per_rank_n
eighbors.csv
-rw-r--r--@ 1 stephenrajva staff 180620 13 Jun 17:09 minimd_avg_hopcount.png
-rw-r--r--@ 1 stephenrajva staff 50625 24 Jun 20:19 minimd_hop_count_vs_size
.png
-rw-r--r--@ 1 stephenrajva staff 53690 24 Jun 20:19 minimd_latency_vs_size.p
ng
-rw-r--r-- 1 stephenrajva staff 430242 13 Jun 17:09 minimd_mesh_simulation_o
utput.pdf
-rw-r--r-- 1 stephenrajva staff 49604 13 Jun 17:09 minimd_mesh_stats.csv
-rw-r--r--@ 1 stephenrajva staff 4424 24 Jun 15:32 minimd_mesh3d_ppn1.py
drwxr-xr-x@ 7 stephenrajva staff 224 24 Jun 15:40 results

===== destination folder (SST_Stimulator MiniMD/Mesh) =====
total 760

```

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drwxr-xr-x 12 stephenrajva staff 384 19 Jun 12:47 .
drwxr-xr-x 9 stephenrajva staff 288 24 Jun 23:24 ..
-rw-r--r-- 1 stephenrajva staff 1654 19 Jun 12:44 extract_mesh_metrics.py
-rw-r--r-- 1 stephenrajva staff 25986 19 Jun 12:47 mesh_latency.png
-rw-r--r--@ 1 stephenrajva staff 26005 19 Jun 12:47 mesh_packets.png
-rw-r--r--@ 1 stephenrajva staff 212 19 Jun 12:44 mesh_summary.csv
-rw-r--r--@ 1 stephenrajva staff 25469 19 Jun 12:47 mesh_throughput.png
-rw-r--r--@ 1 stephenrajva staff 49604 19 Jun 12:33 minimd_mesh_stats.csv
-rw-r--r-- 1 stephenrajva staff 227822 19 Jun 12:39 minimd_mesh_stats64.csv
-rw-r--r-- 1 stephenrajva staff 2872 19 Jun 12:31 minimd_mesh16.py
-rw-r--r-- 1 stephenrajva staff 2873 19 Jun 12:37 minimd_mesh64.py
-rw-r--r-- 1 stephenrajva staff 827 19 Jun 12:47 plot_mesh_scaling.py
stephenrajva@Stephens-MacBook-Air-4 ~ % DEST="/Users/stephenrajva/Downloads/HPC-
Upload 2/SST_Stimulator/MiniMD/Mesh"
SRC="/Users/stephenrajva/Downloads/HPC-Upload 2/miniapps/MiniMD_simulation/mesh3
D"
cp "$SRC/minimd_mesh3d_ppn1.py" "$SRC/analyze_minimd_mesh3d.py" "$DEST/"
mkdir -p "$DEST/results_ppn1"
cp "$SRC/results/*.csv" "$DEST/results_ppn1/" 2>/dev/null
echo "=== Mesh folder now ==="; ls -l "$DEST/"
echo "=== copied PPN=1 CSVs ==="; ls -l "$DEST/results_ppn1/"
=== Mesh folder now ===
total 792
-rw-r--r--@ 1 stephenrajva staff 4131 28 Jun 07:12 analyze_minimd_mesh3d.py
-rw-r--r-- 1 stephenrajva staff 1654 19 Jun 12:44 extract_mesh_metrics.py
-rw-r--r-- 1 stephenrajva staff 25986 19 Jun 12:47 mesh_latency.png
-rw-r--r--@ 1 stephenrajva staff 26005 19 Jun 12:47 mesh_packets.png
-rw-r--r--@ 1 stephenrajva staff 212 19 Jun 12:44 mesh_summary.csv
-rw-r--r--@ 1 stephenrajva staff 25469 19 Jun 12:47 mesh_throughput.png
-rw-r--r--@ 1 stephenrajva staff 49604 19 Jun 12:33 minimd_mesh_stats.csv
-rw-r--r-- 1 stephenrajva staff 227822 19 Jun 12:39 minimd_mesh_stats64.csv
-rw-r--r-- 1 stephenrajva staff 2872 19 Jun 12:31 minimd_mesh16.py
-rw-r--r--@ 1 stephenrajva staff 4424 28 Jun 07:12 minimd_mesh3d_ppn1.py
-rw-r--r-- 1 stephenrajva staff 2873 19 Jun 12:37 minimd_mesh64.py
-rw-r--r-- 1 stephenrajva staff 827 19 Jun 12:47 plot_mesh_scaling.py
drwxr-xr-x 7 stephenrajva staff 224 28 Jun 07:12 results_ppn1
=== copied PPN=1 CSVs ===
total 5448
-rw-r--r--@ 1 stephenrajva staff 22846 28 Jun 07:12 minimd_mesh_2x2x2_ppn1_n
odes8_stats.csv
-rw-r--r-- 1 stephenrajva staff 49604 28 Jun 07:12 minimd_mesh_2x2x4_ppn1_n
odes16_stats.csv
-rw-r--r-- 1 stephenrajva staff 227822 28 Jun 07:12 minimd_mesh_4x4x4_ppn1_n
odes64_stats.csv
-rw-r--r-- 1 stephenrajva staff 471518 28 Jun 07:12 minimd_mesh_4x4x8_ppn1_n
odes128_stats.csv
-rw-r--r-- 1 stephenrajva staff 2006259 28 Jun 07:12 minimd_mesh_8x8x8_ppn1_n
odes512_stats.csv
stephenrajva@Stephens-MacBook-Air-4 ~ % cd "/Users/stephenrajva/Downloads/HPC-Up
load 2/SST_Stimulator/MiniMD/Mesh"
mkdir -p PPN_2
mv ~/Downloads/minimd_mesh3d_ppn2.py ~/Downloads/run_minimd_mesh_sweep_ppn2.sh P
PN_2/
cd PPN_2
chmod +x run_minimd_mesh_sweep_ppn2.sh
export MESH_SHAPE="2x2x2"; export STATS_DIR="./results"
sst minimd_mesh3d_ppn2.py
=====

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Topology : 3D Mesh 2x2x2 - MiniMD
Routers : 8
Nodes per router : 2 (local_ports)
Total nodes : 16
Total MPI ranks : 16
PPN (ranks/node) : 2 <-- should be 2
Stats CSV : ./results/minimd_mesh_2x2x2_ppn2_nodes16_stats.csv

```
=====
180000:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
180000:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
180000:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
180000:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
180000:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
180000:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
180000:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
180000:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
180000:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
180000:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
180000:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
180000:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
180000:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
180000:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
180000:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
180000:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
162220607633:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
162220607633:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
162220607633:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
162220607633:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
162221123891:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
162221123891:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
162221123891:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
162221123891:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
AllPingPong: total time 162223.433 us, loop 325, bufLen 20510, latency 249.575 u
s. bandwidth 0.082180 GB/s
162223613436:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
162223613436:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
162223613436:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
162223613436:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
162224129310:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
162224129310:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
162224129310:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
162224129310:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
318018959042:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
318018959042:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
318018959042:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
318018959042:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
318019542950:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
318019542950:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
318019542950:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
318019542950:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
AllPingPong: total time 155796.843 us, loop 325, bufLen 9000, latency 239.687 us
. bandwidth 0.037549 GB/s
318020456552:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
318020456552:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
318020456552:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
318020456552:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
318021040460:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
318021040460:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
```

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318021040460:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
318021040460:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
480240371383:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
480240371383:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
480240371383:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
480240371383:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
480240887641:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
480240887641:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
480240887641:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
480240887641:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
AllPingPong: total time 162222.921 us, loop 325, bufLen 20510, latency 249.574 u
s. bandwidth 0.082180 GB/s
480243377186:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
480243377186:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
480243377186:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
480243377186:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
480243893060:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
480243893060:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
480243893060:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
480243893060:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
AllPingPong: total time 155796.843 us, loop 325, bufLen 9000, latency 239.687 us
. bandwidth 0.037549 GB/s
Allreduce: ranks 16, loop 40, 1 double(s), latency 2457.854 us
Barrier: ranks 16, loop 15, latency 1398.932 us
Simulation is complete, simulated time: 755.348 ms
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % ./run_minimd_mesh_sweep_ppn2.sh
ls -l results/*.csv
=== MiniMD 3D mesh 2x2x2 (PPN=2) ===
=====
Topology          : 3D Mesh 2x2x2 - MiniMD
Routers           : 8
Nodes per router  : 2 (local_ports)
Total nodes       : 16
Total MPI ranks   : 16
PPN (ranks/node)  : 2  <-- should be 2
Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/M
inimd/Mesh/PPN_2/results/minimd_mesh_2x2x2_ppn2_nodes16_stats.csv
=====
180000:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
180000:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
180000:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
180000:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
180000:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
180000:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
180000:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
180000:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
180000:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
180000:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
180000:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
180000:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
180000:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
180000:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
180000:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
180000:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
162220607633:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
162220607633:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
162220607633:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
162220607633:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16

```



```

162221123891:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
162221123891:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
162221123891:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
162221123891:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
AllPingPong: total time 162223.433 us, loop 325, bufLen 20510, latency 249.575 u
s. bandwidth 0.082180 GB/s
162223613436:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
162223613436:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
162223613436:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
162223613436:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
162224129310:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
162224129310:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
162224129310:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
162224129310:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
318018959042:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
318018959042:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
318018959042:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
318018959042:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
318019542950:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
318019542950:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
318019542950:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
318019542950:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
AllPingPong: total time 155796.843 us, loop 325, bufLen 9000, latency 239.687 us
. bandwidth 0.037549 GB/s
318020456552:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
318020456552:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
318020456552:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
318020456552:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
318021040460:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
318021040460:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
318021040460:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
318021040460:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
480240371383:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
480240371383:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
480240371383:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
480240371383:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
480240887641:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
480240887641:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
480240887641:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
480240887641:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
AllPingPong: total time 162222.921 us, loop 325, bufLen 20510, latency 249.574 u
s. bandwidth 0.082180 GB/s
480243377186:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
480243377186:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
480243377186:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
480243377186:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
480243893060:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
480243893060:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
480243893060:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
480243893060:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
AllPingPong: total time 155796.843 us, loop 325, bufLen 9000, latency 239.687 us
. bandwidth 0.037549 GB/s
Allreduce: ranks 16, loop 40, 1 double(s), latency 2457.854 us
Barrier: ranks 16, loop 15, latency 1398.932 us
Simulation is complete, simulated time: 755.348 ms
=== MiniMD 3D mesh 2x2x4 (PPN=2) ===
=====
Topology                : 3D Mesh 2x2x4 - MiniMD

```

```
Routers          : 16
Nodes per router : 2 (local_ports)
Total nodes      : 32
Total MPI ranks  : 32
PPN (ranks/node) : 2  <-- should be 2
Stats CSV        : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/M
iniMD/Mesh/PPN_2/results/minimd_mesh_2x2x4_ppn2_nodes32_stats.csv
=====
```

```
180000:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=32
180000:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=32
180000:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=32
180000:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=32
180000:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=32
180000:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=32
180000:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=32
180000:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=32
180000:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=32
180000:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=32
180000:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=32
180000:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=32
180000:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=32
180000:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=32
180000:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=32
180000:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=32
180000:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=32
180000:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=32
180000:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=32
180000:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=32
180000:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=32
180000:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=32
180000:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=32
180000:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=32
180000:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=32
180000:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=32
180000:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=32
180000:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=32
180000:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=32
180000:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=32
180000:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=32
180000:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=32
161849408126:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=32
161849408126:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=32
161849408126:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=32
161849408126:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=32
161851118314:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=32
161851118314:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=32
161851118314:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=32
161851118314:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=32
161852352181:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=32
161852352181:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=32
161852352181:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=32
161852352181:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=32
161854062807:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=32
161854062807:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=32
161854062807:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=32
161854062807:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=32
162074970851:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=32
162074970851:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=32
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162074970851:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=32
162074970851:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=32
AllPingPong: total time 162077.735 us, loop 325, bufLen 20510, latency 249.350 u
s. bandwidth 0.082254 GB/s
162077915344:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=32
162077915344:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=32
162077915344:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=32
162077915344:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=32
162346141035:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=32
162346141035:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=32
162346141035:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=32
162346141035:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=32
162349085528:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=32
162349085528:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=32
162349085528:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=32
162349085528:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=32
317576704847:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=32
317576704847:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=32
317576704847:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=32
317576704847:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=32
317578092969:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=32
317578092969:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=32
317578092969:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=32
317578092969:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=32
317578417703:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=32
317578417703:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=32
317578417703:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=32
317578417703:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=32
317579805825:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=32
317579805825:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=32
317579805825:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=32
317579805825:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=32
317802268239:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=32
317802268239:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=32
317802268239:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=32
317802268239:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=32
AllPingPong: total time 155725.741 us, loop 325, bufLen 9000, latency 239.578 us
. bandwidth 0.037566 GB/s
317803656361:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=32
317803656361:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=32
317803656361:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=32
317803656361:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=32
318073438423:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=32
318073438423:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=32
318073438423:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=32
318073438423:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=32
318074826545:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=32
318074826545:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=32
318074826545:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=32
318074826545:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=32
479917996943:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=32
479917996943:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=32
479917996943:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=32
479917996943:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=32
479919711133:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=32
479919711133:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=32
479919711133:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=32
479919711133:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=32

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479920940998:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=32
479920940998:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=32
479920940998:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=32
479920940998:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=32
479921482685:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=32
479921482685:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=32
479921482685:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=32
479921482685:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=32
479922655626:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=32
479922655626:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=32
479922655626:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=32
479922655626:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=32
AllPingPong: total time 162120.771 us, loop 325, bufLen 20510, latency 249.417 u
s. bandwidth 0.082232 GB/s
479924427178:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=32
479924427178:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=32
479924427178:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=32
479924427178:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=32
480414733854:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=32
480414733854:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=32
480414733854:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=32
480414733854:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=32
480417678347:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=32
480417678347:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=32
480417678347:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=32
480417678347:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=32
AllPingPong: total time 155725.741 us, loop 325, bufLen 9000, latency 239.578 us
. bandwidth 0.037566 GB/s
Allreduce: ranks 32, loop 40, 1 double(s), latency 2472.024 us
Barrier: ranks 32, loop 15, latency 1400.828 us
Simulation is complete, simulated time: 755.556 ms
=== MiniMD 3D mesh 4x4x4 (PPN=2) ===
=====
Topology          : 3D Mesh 4x4x4 - MiniMD
Routers           : 64
Nodes per router  : 2 (local_ports)
Total nodes       : 128
Total MPI ranks   : 128
PPN (ranks/node)  : 2 <-- should be 2
Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/M
iniMD/Mesh/PPN_2/results/minimd_mesh_4x4x4_ppn2_nodes128_stats.csv
=====
180000:0:127:EmberEngine:AllPingPong:generate:59: rank=127 size=128
180000:0:126:EmberEngine:AllPingPong:generate:59: rank=126 size=128
180000:0:125:EmberEngine:AllPingPong:generate:59: rank=125 size=128
180000:0:124:EmberEngine:AllPingPong:generate:59: rank=124 size=128
180000:0:123:EmberEngine:AllPingPong:generate:59: rank=123 size=128
180000:0:122:EmberEngine:AllPingPong:generate:59: rank=122 size=128
180000:0:121:EmberEngine:AllPingPong:generate:59: rank=121 size=128
180000:0:120:EmberEngine:AllPingPong:generate:59: rank=120 size=128
180000:0:119:EmberEngine:AllPingPong:generate:59: rank=119 size=128
180000:0:118:EmberEngine:AllPingPong:generate:59: rank=118 size=128
180000:0:117:EmberEngine:AllPingPong:generate:59: rank=117 size=128
180000:0:116:EmberEngine:AllPingPong:generate:59: rank=116 size=128
180000:0:115:EmberEngine:AllPingPong:generate:59: rank=115 size=128
180000:0:114:EmberEngine:AllPingPong:generate:59: rank=114 size=128
180000:0:113:EmberEngine:AllPingPong:generate:59: rank=113 size=128
180000:0:112:EmberEngine:AllPingPong:generate:59: rank=112 size=128

```

[illegible]

[illegible]

[illegible]

161854062807:0:61:EmberEngine:AllPingPong:generate:59: rank=61 size=128
161854062807:0:63:EmberEngine:AllPingPong:generate:59: rank=63 size=128
162074970851:0:64:EmberEngine:AllPingPong:generate:59: rank=64 size=128
162074970851:0:66:EmberEngine:AllPingPong:generate:59: rank=66 size=128
162074970851:0:68:EmberEngine:AllPingPong:generate:59: rank=68 size=128
162074970851:0:70:EmberEngine:AllPingPong:generate:59: rank=70 size=128
162074970851:0:72:EmberEngine:AllPingPong:generate:59: rank=72 size=128
162074970851:0:74:EmberEngine:AllPingPong:generate:59: rank=74 size=128
162074970851:0:76:EmberEngine:AllPingPong:generate:59: rank=76 size=128
162074970851:0:78:EmberEngine:AllPingPong:generate:59: rank=78 size=128
162074970851:0:80:EmberEngine:AllPingPong:generate:59: rank=80 size=128
162074970851:0:82:EmberEngine:AllPingPong:generate:59: rank=82 size=128
162074970851:0:84:EmberEngine:AllPingPong:generate:59: rank=84 size=128
162074970851:0:86:EmberEngine:AllPingPong:generate:59: rank=86 size=128
162074970851:0:88:EmberEngine:AllPingPong:generate:59: rank=88 size=128
162074970851:0:90:EmberEngine:AllPingPong:generate:59: rank=90 size=128
162074970851:0:92:EmberEngine:AllPingPong:generate:59: rank=92 size=128
162074970851:0:94:EmberEngine:AllPingPong:generate:59: rank=94 size=128
AllPingPong: total time 162077.735 us, loop 325, bufLen 20510, latency 249.350 u
s. bandwidth 0.082254 GB/s
162077915344:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=128
162077915344:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=128
162077915344:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=128
162077915344:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=128
162077915344:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=128
162077915344:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=128
162077915344:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=128
162077915344:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=128
162077915344:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=128
162077915344:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=128
162077915344:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=128
162077915344:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=128
162077915344:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=128
162077915344:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=128
162077915344:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=128
162077915344:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=128
162346141035:0:65:EmberEngine:AllPingPong:generate:59: rank=65 size=128
162346141035:0:67:EmberEngine:AllPingPong:generate:59: rank=67 size=128
162346141035:0:69:EmberEngine:AllPingPong:generate:59: rank=69 size=128
162346141035:0:71:EmberEngine:AllPingPong:generate:59: rank=71 size=128
162346141035:0:73:EmberEngine:AllPingPong:generate:59: rank=73 size=128
162346141035:0:75:EmberEngine:AllPingPong:generate:59: rank=75 size=128
162346141035:0:77:EmberEngine:AllPingPong:generate:59: rank=77 size=128
162346141035:0:79:EmberEngine:AllPingPong:generate:59: rank=79 size=128
162346141035:0:81:EmberEngine:AllPingPong:generate:59: rank=81 size=128
162346141035:0:83:EmberEngine:AllPingPong:generate:59: rank=83 size=128
162346141035:0:85:EmberEngine:AllPingPong:generate:59: rank=85 size=128
162346141035:0:87:EmberEngine:AllPingPong:generate:59: rank=87 size=128
162346141035:0:89:EmberEngine:AllPingPong:generate:59: rank=89 size=128
162346141035:0:91:EmberEngine:AllPingPong:generate:59: rank=91 size=128
162346141035:0:93:EmberEngine:AllPingPong:generate:59: rank=93 size=128
162346141035:0:95:EmberEngine:AllPingPong:generate:59: rank=95 size=128
162349085528:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=128
162349085528:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=128
162349085528:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=128
162349085528:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=128
162349085528:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=128
162349085528:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=128

[illegible]

317579805825:0:33:EmberEngine:AllPingPong:generate:59: rank=33 size=128
317579805825:0:35:EmberEngine:AllPingPong:generate:59: rank=35 size=128
317579805825:0:37:EmberEngine:AllPingPong:generate:59: rank=37 size=128
317579805825:0:39:EmberEngine:AllPingPong:generate:59: rank=39 size=128
317579805825:0:41:EmberEngine:AllPingPong:generate:59: rank=41 size=128
317579805825:0:43:EmberEngine:AllPingPong:generate:59: rank=43 size=128
317579805825:0:45:EmberEngine:AllPingPong:generate:59: rank=45 size=128
317579805825:0:47:EmberEngine:AllPingPong:generate:59: rank=47 size=128
317579805825:0:49:EmberEngine:AllPingPong:generate:59: rank=49 size=128
317579805825:0:51:EmberEngine:AllPingPong:generate:59: rank=51 size=128
317579805825:0:53:EmberEngine:AllPingPong:generate:59: rank=53 size=128
317579805825:0:55:EmberEngine:AllPingPong:generate:59: rank=55 size=128
317579805825:0:57:EmberEngine:AllPingPong:generate:59: rank=57 size=128
317579805825:0:59:EmberEngine:AllPingPong:generate:59: rank=59 size=128
317579805825:0:61:EmberEngine:AllPingPong:generate:59: rank=61 size=128
317579805825:0:63:EmberEngine:AllPingPong:generate:59: rank=63 size=128
317802268239:0:64:EmberEngine:AllPingPong:generate:59: rank=64 size=128
317802268239:0:66:EmberEngine:AllPingPong:generate:59: rank=66 size=128
317802268239:0:68:EmberEngine:AllPingPong:generate:59: rank=68 size=128
317802268239:0:70:EmberEngine:AllPingPong:generate:59: rank=70 size=128
317802268239:0:72:EmberEngine:AllPingPong:generate:59: rank=72 size=128
317802268239:0:74:EmberEngine:AllPingPong:generate:59: rank=74 size=128
317802268239:0:76:EmberEngine:AllPingPong:generate:59: rank=76 size=128
317802268239:0:78:EmberEngine:AllPingPong:generate:59: rank=78 size=128
317802268239:0:80:EmberEngine:AllPingPong:generate:59: rank=80 size=128
317802268239:0:82:EmberEngine:AllPingPong:generate:59: rank=82 size=128
317802268239:0:84:EmberEngine:AllPingPong:generate:59: rank=84 size=128
317802268239:0:86:EmberEngine:AllPingPong:generate:59: rank=86 size=128
317802268239:0:88:EmberEngine:AllPingPong:generate:59: rank=88 size=128
317802268239:0:90:EmberEngine:AllPingPong:generate:59: rank=90 size=128
317802268239:0:92:EmberEngine:AllPingPong:generate:59: rank=92 size=128
317802268239:0:94:EmberEngine:AllPingPong:generate:59: rank=94 size=128
AllPingPong: total time 155725.741 us, loop 325, bufLen 9000, latency 239.578 us
. bandwidth 0.037566 GB/s
317803656361:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=128
317803656361:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=128
317803656361:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=128
317803656361:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=128
317803656361:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=128
317803656361:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=128
317803656361:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=128
317803656361:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=128
317803656361:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=128
317803656361:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=128
317803656361:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=128
317803656361:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=128
317803656361:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=128
317803656361:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=128
317803656361:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=128
317803656361:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=128
318073438423:0:65:EmberEngine:AllPingPong:generate:59: rank=65 size=128
318073438423:0:67:EmberEngine:AllPingPong:generate:59: rank=67 size=128
318073438423:0:69:EmberEngine:AllPingPong:generate:59: rank=69 size=128
318073438423:0:71:EmberEngine:AllPingPong:generate:59: rank=71 size=128
318073438423:0:73:EmberEngine:AllPingPong:generate:59: rank=73 size=128
318073438423:0:75:EmberEngine:AllPingPong:generate:59: rank=75 size=128
318073438423:0:77:EmberEngine:AllPingPong:generate:59: rank=77 size=128
318073438423:0:79:EmberEngine:AllPingPong:generate:59: rank=79 size=128

[illegible]

479920940998:0:36:EmberEngine:AllPingPong:generate:59: rank=36 size=128
479920940998:0:38:EmberEngine:AllPingPong:generate:59: rank=38 size=128
479920940998:0:40:EmberEngine:AllPingPong:generate:59: rank=40 size=128
479920940998:0:42:EmberEngine:AllPingPong:generate:59: rank=42 size=128
479920940998:0:44:EmberEngine:AllPingPong:generate:59: rank=44 size=128
479920940998:0:46:EmberEngine:AllPingPong:generate:59: rank=46 size=128
479920940998:0:48:EmberEngine:AllPingPong:generate:59: rank=48 size=128
479920940998:0:50:EmberEngine:AllPingPong:generate:59: rank=50 size=128
479920940998:0:52:EmberEngine:AllPingPong:generate:59: rank=52 size=128
479920940998:0:54:EmberEngine:AllPingPong:generate:59: rank=54 size=128
479920940998:0:56:EmberEngine:AllPingPong:generate:59: rank=56 size=128
479920940998:0:58:EmberEngine:AllPingPong:generate:59: rank=58 size=128
479920940998:0:60:EmberEngine:AllPingPong:generate:59: rank=60 size=128
479920940998:0:62:EmberEngine:AllPingPong:generate:59: rank=62 size=128
479921482685:0:64:EmberEngine:AllPingPong:generate:59: rank=64 size=128
479921482685:0:66:EmberEngine:AllPingPong:generate:59: rank=66 size=128
479921482685:0:68:EmberEngine:AllPingPong:generate:59: rank=68 size=128
479921482685:0:70:EmberEngine:AllPingPong:generate:59: rank=70 size=128
479921482685:0:72:EmberEngine:AllPingPong:generate:59: rank=72 size=128
479921482685:0:74:EmberEngine:AllPingPong:generate:59: rank=74 size=128
479921482685:0:76:EmberEngine:AllPingPong:generate:59: rank=76 size=128
479921482685:0:78:EmberEngine:AllPingPong:generate:59: rank=78 size=128
479921482685:0:80:EmberEngine:AllPingPong:generate:59: rank=80 size=128
479921482685:0:82:EmberEngine:AllPingPong:generate:59: rank=82 size=128
479921482685:0:84:EmberEngine:AllPingPong:generate:59: rank=84 size=128
479921482685:0:86:EmberEngine:AllPingPong:generate:59: rank=86 size=128
479921482685:0:88:EmberEngine:AllPingPong:generate:59: rank=88 size=128
479921482685:0:90:EmberEngine:AllPingPong:generate:59: rank=90 size=128
479921482685:0:92:EmberEngine:AllPingPong:generate:59: rank=92 size=128
479921482685:0:94:EmberEngine:AllPingPong:generate:59: rank=94 size=128
479922655626:0:33:EmberEngine:AllPingPong:generate:59: rank=33 size=128
479922655626:0:35:EmberEngine:AllPingPong:generate:59: rank=35 size=128
479922655626:0:37:EmberEngine:AllPingPong:generate:59: rank=37 size=128
479922655626:0:39:EmberEngine:AllPingPong:generate:59: rank=39 size=128
479922655626:0:41:EmberEngine:AllPingPong:generate:59: rank=41 size=128
479922655626:0:43:EmberEngine:AllPingPong:generate:59: rank=43 size=128
479922655626:0:45:EmberEngine:AllPingPong:generate:59: rank=45 size=128
479922655626:0:47:EmberEngine:AllPingPong:generate:59: rank=47 size=128
479922655626:0:49:EmberEngine:AllPingPong:generate:59: rank=49 size=128
479922655626:0:51:EmberEngine:AllPingPong:generate:59: rank=51 size=128
479922655626:0:53:EmberEngine:AllPingPong:generate:59: rank=53 size=128
479922655626:0:55:EmberEngine:AllPingPong:generate:59: rank=55 size=128
479922655626:0:57:EmberEngine:AllPingPong:generate:59: rank=57 size=128
479922655626:0:59:EmberEngine:AllPingPong:generate:59: rank=59 size=128
479922655626:0:61:EmberEngine:AllPingPong:generate:59: rank=61 size=128
479922655626:0:63:EmberEngine:AllPingPong:generate:59: rank=63 size=128
AllPingPong: total time 162120.771 us, loop 325, bufLen 20510, latency 249.417 u
s. bandwidth 0.082232 GB/s
479924427178:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=128
479924427178:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=128
479924427178:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=128
479924427178:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=128
479924427178:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=128
479924427178:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=128
479924427178:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=128
479924427178:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=128
479924427178:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=128
479924427178:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=128

```

479924427178:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=128
479924427178:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=128
479924427178:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=128
479924427178:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=128
479924427178:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=128
479924427178:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=128
480414733854:0:65:EmberEngine:AllPingPong:generate:59: rank=65 size=128
480414733854:0:67:EmberEngine:AllPingPong:generate:59: rank=67 size=128
480414733854:0:69:EmberEngine:AllPingPong:generate:59: rank=69 size=128
480414733854:0:71:EmberEngine:AllPingPong:generate:59: rank=71 size=128
480414733854:0:73:EmberEngine:AllPingPong:generate:59: rank=73 size=128
480414733854:0:75:EmberEngine:AllPingPong:generate:59: rank=75 size=128
480414733854:0:77:EmberEngine:AllPingPong:generate:59: rank=77 size=128
480414733854:0:79:EmberEngine:AllPingPong:generate:59: rank=79 size=128
480414733854:0:81:EmberEngine:AllPingPong:generate:59: rank=81 size=128
480414733854:0:83:EmberEngine:AllPingPong:generate:59: rank=83 size=128
480414733854:0:85:EmberEngine:AllPingPong:generate:59: rank=85 size=128
480414733854:0:87:EmberEngine:AllPingPong:generate:59: rank=87 size=128
480414733854:0:89:EmberEngine:AllPingPong:generate:59: rank=89 size=128
480414733854:0:91:EmberEngine:AllPingPong:generate:59: rank=91 size=128
480414733854:0:93:EmberEngine:AllPingPong:generate:59: rank=93 size=128
480414733854:0:95:EmberEngine:AllPingPong:generate:59: rank=95 size=128
480417678347:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=128
480417678347:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=128
480417678347:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=128
480417678347:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=128
480417678347:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=128
480417678347:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=128
480417678347:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=128
480417678347:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=128
480417678347:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=128
480417678347:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=128
480417678347:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=128
480417678347:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=128
480417678347:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=128
480417678347:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=128
480417678347:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=128
480417678347:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=128
AllPingPong: total time 155725.741 us, loop 325, bufLen 9000, latency 239.578 us
. bandwidth 0.037566 GB/s
Allreduce: ranks 128, loop 40, 1double(s), latency 2476.264 us
Barrier: ranks 128, loop 15, latency 1405.115 us
Simulation is complete, simulated time: 755.797 ms
=== MiniMD 3D mesh 4x4x8 (PPN=2) ===
=====
Topology          : 3D Mesh 4x4x8 - MiniMD
Routers           : 128
Nodes per router  : 2 (local_ports)
Total nodes       : 256
Total MPI ranks   : 256
PPN (ranks/node)  : 2  <-- should be 2
Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/M
iniMD/Mesh/PPN_2/results/minimd_mesh_4x4x8_ppn2_nodes256_stats.csv
=====
180000:0:255:EmberEngine:AllPingPong:generate:59: rank=255 size=256
180000:0:254:EmberEngine:AllPingPong:generate:59: rank=254 size=256
180000:0:253:EmberEngine:AllPingPong:generate:59: rank=253 size=256
180000:0:252:EmberEngine:AllPingPong:generate:59: rank=252 size=256

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[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

162602152483:0:41:EmberEngine:AllPingPong:generate:59: rank=41 size=256
162602152483:0:43:EmberEngine:AllPingPong:generate:59: rank=43 size=256
162602152483:0:45:EmberEngine:AllPingPong:generate:59: rank=45 size=256
162602152483:0:47:EmberEngine:AllPingPong:generate:59: rank=47 size=256
162602152483:0:49:EmberEngine:AllPingPong:generate:59: rank=49 size=256
162602152483:0:51:EmberEngine:AllPingPong:generate:59: rank=51 size=256
162602152483:0:53:EmberEngine:AllPingPong:generate:59: rank=53 size=256
162602152483:0:55:EmberEngine:AllPingPong:generate:59: rank=55 size=256
162602152483:0:57:EmberEngine:AllPingPong:generate:59: rank=57 size=256
162602152483:0:59:EmberEngine:AllPingPong:generate:59: rank=59 size=256
162602152483:0:61:EmberEngine:AllPingPong:generate:59: rank=61 size=256
162602152483:0:63:EmberEngine:AllPingPong:generate:59: rank=63 size=256
AllPingPong: total time 162602.549 us, loop 325, bufLen 20510, latency 250.158 u
s. bandwidth 0.081988 GB/s
162602729849:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=256
162602729849:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=256
162602729849:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=256
162602729849:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=256
162602729849:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=256
162602729849:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=256
162602729849:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=256
162602729849:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=256
162602729849:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=256
162602729849:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=256
162602729849:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=256
162602729849:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=256
162602729849:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=256
162602729849:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=256
162602729849:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=256
162602729849:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=256
162604440475:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=256
162604440475:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=256
162604440475:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=256
162604440475:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=256
162604440475:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=256
162604440475:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=256
162604440475:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=256
162604440475:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=256
162604440475:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=256
162604440475:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=256
162604440475:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=256
162604440475:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=256
162604440475:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=256
162604440475:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=256
162604440475:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=256
162604440475:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=256
318392926754:0:224:EmberEngine:AllPingPong:generate:59: rank=224 size=256
318392926754:0:226:EmberEngine:AllPingPong:generate:59: rank=226 size=256
318392926754:0:228:EmberEngine:AllPingPong:generate:59: rank=228 size=256
318392926754:0:230:EmberEngine:AllPingPong:generate:59: rank=230 size=256
318392926754:0:232:EmberEngine:AllPingPong:generate:59: rank=232 size=256
318392926754:0:234:EmberEngine:AllPingPong:generate:59: rank=234 size=256
318392926754:0:236:EmberEngine:AllPingPong:generate:59: rank=236 size=256
318392926754:0:238:EmberEngine:AllPingPong:generate:59: rank=238 size=256
318392926754:0:240:EmberEngine:AllPingPong:generate:59: rank=240 size=256
318392926754:0:242:EmberEngine:AllPingPong:generate:59: rank=242 size=256
318392926754:0:244:EmberEngine:AllPingPong:generate:59: rank=244 size=256
318392926754:0:246:EmberEngine:AllPingPong:generate:59: rank=246 size=256

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318572819450:0:148:EmberEngine:AllPingPong:generate:59: rank=148 size=256
318572819450:0:150:EmberEngine:AllPingPong:generate:59: rank=150 size=256
318572819450:0:152:EmberEngine:AllPingPong:generate:59: rank=152 size=256
318572819450:0:154:EmberEngine:AllPingPong:generate:59: rank=154 size=256
318572819450:0:156:EmberEngine:AllPingPong:generate:59: rank=156 size=256
318572819450:0:158:EmberEngine:AllPingPong:generate:59: rank=158 size=256
318573574366:0:129:EmberEngine:AllPingPong:generate:59: rank=129 size=256
318573574366:0:131:EmberEngine:AllPingPong:generate:59: rank=131 size=256
318573574366:0:133:EmberEngine:AllPingPong:generate:59: rank=133 size=256
318573574366:0:135:EmberEngine:AllPingPong:generate:59: rank=135 size=256
318573574366:0:137:EmberEngine:AllPingPong:generate:59: rank=137 size=256
318573574366:0:139:EmberEngine:AllPingPong:generate:59: rank=139 size=256
318573574366:0:141:EmberEngine:AllPingPong:generate:59: rank=141 size=256
318573574366:0:143:EmberEngine:AllPingPong:generate:59: rank=143 size=256
318573574366:0:145:EmberEngine:AllPingPong:generate:59: rank=145 size=256
318573574366:0:147:EmberEngine:AllPingPong:generate:59: rank=147 size=256
318573574366:0:149:EmberEngine:AllPingPong:generate:59: rank=149 size=256
318573574366:0:151:EmberEngine:AllPingPong:generate:59: rank=151 size=256
318573574366:0:153:EmberEngine:AllPingPong:generate:59: rank=153 size=256
318573574366:0:155:EmberEngine:AllPingPong:generate:59: rank=155 size=256
318573574366:0:157:EmberEngine:AllPingPong:generate:59: rank=157 size=256
318573574366:0:159:EmberEngine:AllPingPong:generate:59: rank=159 size=256
318574149260:0:33:EmberEngine:AllPingPong:generate:59: rank=33 size=256
318574149260:0:35:EmberEngine:AllPingPong:generate:59: rank=35 size=256
318574149260:0:37:EmberEngine:AllPingPong:generate:59: rank=37 size=256
318574149260:0:39:EmberEngine:AllPingPong:generate:59: rank=39 size=256
318574149260:0:41:EmberEngine:AllPingPong:generate:59: rank=41 size=256
318574149260:0:43:EmberEngine:AllPingPong:generate:59: rank=43 size=256
318574149260:0:45:EmberEngine:AllPingPong:generate:59: rank=45 size=256
318574149260:0:47:EmberEngine:AllPingPong:generate:59: rank=47 size=256
318574149260:0:49:EmberEngine:AllPingPong:generate:59: rank=49 size=256
318574149260:0:51:EmberEngine:AllPingPong:generate:59: rank=51 size=256
318574149260:0:53:EmberEngine:AllPingPong:generate:59: rank=53 size=256
318574149260:0:55:EmberEngine:AllPingPong:generate:59: rank=55 size=256
318574149260:0:57:EmberEngine:AllPingPong:generate:59: rank=57 size=256
318574149260:0:59:EmberEngine:AllPingPong:generate:59: rank=59 size=256
318574149260:0:61:EmberEngine:AllPingPong:generate:59: rank=61 size=256
318574149260:0:63:EmberEngine:AllPingPong:generate:59: rank=63 size=256
AllPingPong: total time 155971.894 us, loop 325, bufLen 9000, latency 239.957 us
. bandwidth 0.037507 GB/s
318574623908:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=256
318574623908:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=256
318574623908:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=256
318574623908:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=256
318574623908:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=256
318574623908:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=256
318574623908:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=256
318574623908:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=256
318574623908:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=256
318574623908:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=256
318574623908:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=256
318574623908:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=256
318574623908:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=256
318574623908:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=256
318574623908:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=256
318574623908:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=256
318575378696:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=256
318575378696:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=256

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481037341002:0:161:EmberEngine:AllPingPong:generate:59: rank=161 size=256
481037341002:0:163:EmberEngine:AllPingPong:generate:59: rank=163 size=256
481037341002:0:165:EmberEngine:AllPingPong:generate:59: rank=165 size=256
481037341002:0:167:EmberEngine:AllPingPong:generate:59: rank=167 size=256
481037341002:0:169:EmberEngine:AllPingPong:generate:59: rank=169 size=256
481037341002:0:171:EmberEngine:AllPingPong:generate:59: rank=171 size=256
481037341002:0:173:EmberEngine:AllPingPong:generate:59: rank=173 size=256
481037341002:0:175:EmberEngine:AllPingPong:generate:59: rank=175 size=256
481037341002:0:177:EmberEngine:AllPingPong:generate:59: rank=177 size=256
481037341002:0:179:EmberEngine:AllPingPong:generate:59: rank=179 size=256
481037341002:0:181:EmberEngine:AllPingPong:generate:59: rank=181 size=256
481037341002:0:183:EmberEngine:AllPingPong:generate:59: rank=183 size=256
481037341002:0:185:EmberEngine:AllPingPong:generate:59: rank=185 size=256
481037341002:0:187:EmberEngine:AllPingPong:generate:59: rank=187 size=256
481037341002:0:189:EmberEngine:AllPingPong:generate:59: rank=189 size=256
481037341002:0:191:EmberEngine:AllPingPong:generate:59: rank=191 size=256
481037918624:0:128:EmberEngine:AllPingPong:generate:59: rank=128 size=256
481037918624:0:130:EmberEngine:AllPingPong:generate:59: rank=130 size=256
481037918624:0:132:EmberEngine:AllPingPong:generate:59: rank=132 size=256
481037918624:0:134:EmberEngine:AllPingPong:generate:59: rank=134 size=256
481037918624:0:136:EmberEngine:AllPingPong:generate:59: rank=136 size=256
481037918624:0:138:EmberEngine:AllPingPong:generate:59: rank=138 size=256
481037918624:0:140:EmberEngine:AllPingPong:generate:59: rank=140 size=256
481037918624:0:142:EmberEngine:AllPingPong:generate:59: rank=142 size=256
481037918624:0:144:EmberEngine:AllPingPong:generate:59: rank=144 size=256
481037918624:0:146:EmberEngine:AllPingPong:generate:59: rank=146 size=256
481037918624:0:148:EmberEngine:AllPingPong:generate:59: rank=148 size=256
481037918624:0:150:EmberEngine:AllPingPong:generate:59: rank=150 size=256
481037918624:0:152:EmberEngine:AllPingPong:generate:59: rank=152 size=256
481037918624:0:154:EmberEngine:AllPingPong:generate:59: rank=154 size=256
481037918624:0:156:EmberEngine:AllPingPong:generate:59: rank=156 size=256
481037918624:0:158:EmberEngine:AllPingPong:generate:59: rank=158 size=256
481040162585:0:33:EmberEngine:AllPingPong:generate:59: rank=33 size=256
481040162585:0:35:EmberEngine:AllPingPong:generate:59: rank=35 size=256
481040162585:0:37:EmberEngine:AllPingPong:generate:59: rank=37 size=256
481040162585:0:39:EmberEngine:AllPingPong:generate:59: rank=39 size=256
481040162585:0:41:EmberEngine:AllPingPong:generate:59: rank=41 size=256
481040162585:0:43:EmberEngine:AllPingPong:generate:59: rank=43 size=256
481040162585:0:45:EmberEngine:AllPingPong:generate:59: rank=45 size=256
481040162585:0:47:EmberEngine:AllPingPong:generate:59: rank=47 size=256
481040162585:0:49:EmberEngine:AllPingPong:generate:59: rank=49 size=256
481040162585:0:51:EmberEngine:AllPingPong:generate:59: rank=51 size=256
481040162585:0:53:EmberEngine:AllPingPong:generate:59: rank=53 size=256
481040162585:0:55:EmberEngine:AllPingPong:generate:59: rank=55 size=256
481040162585:0:57:EmberEngine:AllPingPong:generate:59: rank=57 size=256
481040162585:0:59:EmberEngine:AllPingPong:generate:59: rank=59 size=256
481040162585:0:61:EmberEngine:AllPingPong:generate:59: rank=61 size=256
481040162585:0:63:EmberEngine:AllPingPong:generate:59: rank=63 size=256
AllPingPong: total time 162466.116 us, loop 325, bufLen 20510, latency 249.948 u
s. bandwidth 0.082057 GB/s
481040739951:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=256
481040739951:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=256
481040739951:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=256
481040739951:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=256
481040739951:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=256
481040739951:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=256
481040739951:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=256
481040739951:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=256

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481040739951:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=256
481040739951:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=256
481040739951:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=256
481040739951:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=256
481040739951:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=256
481040739951:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=256
481040739951:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=256
481040739951:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=256
481152751345:0:129:EmberEngine:AllPingPong:generate:59: rank=129 size=256
481152751345:0:131:EmberEngine:AllPingPong:generate:59: rank=131 size=256
481152751345:0:133:EmberEngine:AllPingPong:generate:59: rank=133 size=256
481152751345:0:135:EmberEngine:AllPingPong:generate:59: rank=135 size=256
481152751345:0:137:EmberEngine:AllPingPong:generate:59: rank=137 size=256
481152751345:0:139:EmberEngine:AllPingPong:generate:59: rank=139 size=256
481152751345:0:141:EmberEngine:AllPingPong:generate:59: rank=141 size=256
481152751345:0:143:EmberEngine:AllPingPong:generate:59: rank=143 size=256
481152751345:0:145:EmberEngine:AllPingPong:generate:59: rank=145 size=256
481152751345:0:147:EmberEngine:AllPingPong:generate:59: rank=147 size=256
481152751345:0:149:EmberEngine:AllPingPong:generate:59: rank=149 size=256
481152751345:0:151:EmberEngine:AllPingPong:generate:59: rank=151 size=256
481152751345:0:153:EmberEngine:AllPingPong:generate:59: rank=153 size=256
481152751345:0:155:EmberEngine:AllPingPong:generate:59: rank=155 size=256
481152751345:0:157:EmberEngine:AllPingPong:generate:59: rank=157 size=256
481152751345:0:159:EmberEngine:AllPingPong:generate:59: rank=159 size=256
481155573110:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=256
481155573110:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=256
481155573110:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=256
481155573110:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=256
481155573110:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=256
481155573110:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=256
481155573110:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=256
481155573110:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=256
481155573110:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=256
481155573110:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=256
481155573110:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=256
481155573110:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=256
481155573110:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=256
481155573110:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=256
481155573110:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=256
481155573110:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=256
AllPingPong: total time 155968.501 us, loop 325, bufLen 9000, latency 239.952 us
. bandwidth 0.037508 GB/s
Allreduce: ranks 256, loop 40, 1 double(s), latency 2466.623 us
Barrier: ranks 256, loop 15, latency 1407.823 us
Simulation is complete, simulated time: 756.815 ms
=== MiniMD 3D mesh 8x8x8 (PPN=2) ===
=====
Topology          : 3D Mesh 8x8x8 - MiniMD
Routers           : 512
Nodes per router  : 2 (local_ports)
Total nodes       : 1024
Total MPI ranks   : 1024
PPN (ranks/node)  : 2 <-- should be 2
Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/M
iniMD/Mesh/PPN_2/results/minimd_mesh_8x8x8_ppn2_nodes1024_stats.csv
=====
180000:0:1023:EmberEngine:AllPingPong:generate:59: rank=1023 size=1024
180000:0:1022:EmberEngine:AllPingPong:generate:59: rank=1022 size=1024

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162602152483:0:249:EmberEngine:AllPingPong:generate:59: rank=249 size=1024
162602152483:0:251:EmberEngine:AllPingPong:generate:59: rank=251 size=1024
162602152483:0:253:EmberEngine:AllPingPong:generate:59: rank=253 size=1024
162602152483:0:255:EmberEngine:AllPingPong:generate:59: rank=255 size=1024
AllPingPong: total time 162602.549 us, loop 325, bufLen 20510, latency 250.158 u
s. bandwidth 0.081988 GB/s
162602729849:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=1024
162602729849:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=1024
162602729849:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=1024
162602729849:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=1024
162602729849:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=1024
162602729849:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=1024
162602729849:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=1024
162602729849:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=1024
162602729849:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=1024
162602729849:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=1024
162602729849:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=1024
162602729849:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=1024
162602729849:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=1024
162602729849:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=1024
162602729849:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=1024
162602729849:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=1024
162602729849:0:32:EmberEngine:AllPingPong:generate:59: rank=32 size=1024
162602729849:0:34:EmberEngine:AllPingPong:generate:59: rank=34 size=1024
162602729849:0:36:EmberEngine:AllPingPong:generate:59: rank=36 size=1024
162602729849:0:38:EmberEngine:AllPingPong:generate:59: rank=38 size=1024
162602729849:0:40:EmberEngine:AllPingPong:generate:59: rank=40 size=1024
162602729849:0:42:EmberEngine:AllPingPong:generate:59: rank=42 size=1024
162602729849:0:44:EmberEngine:AllPingPong:generate:59: rank=44 size=1024
162602729849:0:46:EmberEngine:AllPingPong:generate:59: rank=46 size=1024
162602729849:0:48:EmberEngine:AllPingPong:generate:59: rank=48 size=1024
162602729849:0:50:EmberEngine:AllPingPong:generate:59: rank=50 size=1024
162602729849:0:52:EmberEngine:AllPingPong:generate:59: rank=52 size=1024
162602729849:0:54:EmberEngine:AllPingPong:generate:59: rank=54 size=1024
162602729849:0:56:EmberEngine:AllPingPong:generate:59: rank=56 size=1024
162602729849:0:58:EmberEngine:AllPingPong:generate:59: rank=58 size=1024
162602729849:0:60:EmberEngine:AllPingPong:generate:59: rank=60 size=1024
162602729849:0:62:EmberEngine:AllPingPong:generate:59: rank=62 size=1024
162602729849:0:64:EmberEngine:AllPingPong:generate:59: rank=64 size=1024
162602729849:0:66:EmberEngine:AllPingPong:generate:59: rank=66 size=1024
162602729849:0:68:EmberEngine:AllPingPong:generate:59: rank=68 size=1024
162602729849:0:70:EmberEngine:AllPingPong:generate:59: rank=70 size=1024
162602729849:0:72:EmberEngine:AllPingPong:generate:59: rank=72 size=1024
162602729849:0:74:EmberEngine:AllPingPong:generate:59: rank=74 size=1024
162602729849:0:76:EmberEngine:AllPingPong:generate:59: rank=76 size=1024
162602729849:0:78:EmberEngine:AllPingPong:generate:59: rank=78 size=1024
162602729849:0:80:EmberEngine:AllPingPong:generate:59: rank=80 size=1024
162602729849:0:82:EmberEngine:AllPingPong:generate:59: rank=82 size=1024
162602729849:0:84:EmberEngine:AllPingPong:generate:59: rank=84 size=1024
162602729849:0:86:EmberEngine:AllPingPong:generate:59: rank=86 size=1024
162602729849:0:88:EmberEngine:AllPingPong:generate:59: rank=88 size=1024
162602729849:0:90:EmberEngine:AllPingPong:generate:59: rank=90 size=1024
162602729849:0:92:EmberEngine:AllPingPong:generate:59: rank=92 size=1024
162602729849:0:94:EmberEngine:AllPingPong:generate:59: rank=94 size=1024
162602729849:0:96:EmberEngine:AllPingPong:generate:59: rank=96 size=1024
162602729849:0:98:EmberEngine:AllPingPong:generate:59: rank=98 size=1024
162602729849:0:100:EmberEngine:AllPingPong:generate:59: rank=100 size=1024
162602729849:0:102:EmberEngine:AllPingPong:generate:59: rank=102 size=1024

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481040162585:0:217:EmberEngine:AllPingPong:generate:59: rank=217 size=1024
481040162585:0:219:EmberEngine:AllPingPong:generate:59: rank=219 size=1024
481040162585:0:221:EmberEngine:AllPingPong:generate:59: rank=221 size=1024
481040162585:0:223:EmberEngine:AllPingPong:generate:59: rank=223 size=1024
481040162585:0:225:EmberEngine:AllPingPong:generate:59: rank=225 size=1024
481040162585:0:227:EmberEngine:AllPingPong:generate:59: rank=227 size=1024
481040162585:0:229:EmberEngine:AllPingPong:generate:59: rank=229 size=1024
481040162585:0:231:EmberEngine:AllPingPong:generate:59: rank=231 size=1024
481040162585:0:233:EmberEngine:AllPingPong:generate:59: rank=233 size=1024
481040162585:0:235:EmberEngine:AllPingPong:generate:59: rank=235 size=1024
481040162585:0:237:EmberEngine:AllPingPong:generate:59: rank=237 size=1024
481040162585:0:239:EmberEngine:AllPingPong:generate:59: rank=239 size=1024
481040162585:0:241:EmberEngine:AllPingPong:generate:59: rank=241 size=1024
481040162585:0:243:EmberEngine:AllPingPong:generate:59: rank=243 size=1024
481040162585:0:245:EmberEngine:AllPingPong:generate:59: rank=245 size=1024
481040162585:0:247:EmberEngine:AllPingPong:generate:59: rank=247 size=1024
481040162585:0:249:EmberEngine:AllPingPong:generate:59: rank=249 size=1024
481040162585:0:251:EmberEngine:AllPingPong:generate:59: rank=251 size=1024
481040162585:0:253:EmberEngine:AllPingPong:generate:59: rank=253 size=1024
481040162585:0:255:EmberEngine:AllPingPong:generate:59: rank=255 size=1024
AllPingPong: total time 162466.116 us, loop 325, bufLen 20510, latency 249.948 u
s. bandwidth 0.082057 GB/s
481040739951:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=1024
481040739951:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=1024
481040739951:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=1024
481040739951:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=1024
481040739951:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=1024
481040739951:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=1024
481040739951:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=1024
481040739951:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=1024
481040739951:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=1024
481040739951:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=1024
481040739951:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=1024
481040739951:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=1024
481040739951:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=1024
481040739951:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=1024
481040739951:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=1024
481040739951:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=1024
481040739951:0:32:EmberEngine:AllPingPong:generate:59: rank=32 size=1024
481040739951:0:34:EmberEngine:AllPingPong:generate:59: rank=34 size=1024
481040739951:0:36:EmberEngine:AllPingPong:generate:59: rank=36 size=1024
481040739951:0:38:EmberEngine:AllPingPong:generate:59: rank=38 size=1024
481040739951:0:40:EmberEngine:AllPingPong:generate:59: rank=40 size=1024
481040739951:0:42:EmberEngine:AllPingPong:generate:59: rank=42 size=1024
481040739951:0:44:EmberEngine:AllPingPong:generate:59: rank=44 size=1024
481040739951:0:46:EmberEngine:AllPingPong:generate:59: rank=46 size=1024
481040739951:0:48:EmberEngine:AllPingPong:generate:59: rank=48 size=1024
481040739951:0:50:EmberEngine:AllPingPong:generate:59: rank=50 size=1024
481040739951:0:52:EmberEngine:AllPingPong:generate:59: rank=52 size=1024
481040739951:0:54:EmberEngine:AllPingPong:generate:59: rank=54 size=1024
481040739951:0:56:EmberEngine:AllPingPong:generate:59: rank=56 size=1024
481040739951:0:58:EmberEngine:AllPingPong:generate:59: rank=58 size=1024
481040739951:0:60:EmberEngine:AllPingPong:generate:59: rank=60 size=1024
481040739951:0:62:EmberEngine:AllPingPong:generate:59: rank=62 size=1024
481040739951:0:64:EmberEngine:AllPingPong:generate:59: rank=64 size=1024
481040739951:0:66:EmberEngine:AllPingPong:generate:59: rank=66 size=1024
481040739951:0:68:EmberEngine:AllPingPong:generate:59: rank=68 size=1024
481040739951:0:70:EmberEngine:AllPingPong:generate:59: rank=70 size=1024

[illegible]

[illegible]